

Brock Biology of Microorganisms, 15e (Madigan et al.)
Chapter 28 Clinical Microbiology and Immunology

28.1 Multiple Choice Questions

- 1) Laboratories that work with extremely low risk pathogens are classified as
A) BSL-1.
B) BSL-2.
C) BSL-3.
D) BSL-4.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.1

- 2) Which of the following must be accomplished FIRST in order to conduct RT-PCR?
A) Complementary DNA must be synthesized from double-stranded DNA.
B) Complementary DNA must be synthesized from RNA.
C) Complementary DNA must be synthesized from proteins.
D) Complementary DNA must be synthesized from single-stranded DNA.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.8

- 3) _____ infections are among the most common nosocomial infections.
A) Lymphatic
B) Reproductive tract
C) Skin
D) Respiratory tract

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.3

- 4) MacConkey agar and eosin-methylene blue (EMB) agar are _____ media.
A) selective but not differential
B) differential but not selective
C) both selective and differential
D) neither selective nor differential

Answer: C

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 28.3

- 5) An Etest[®] is a clinical tool used for the determination of
- A) enteric presence.
 - B) *Escherichia coli* O157:H7 presence.
 - C) the minimum inhibitory concentration (MIC) value of an antimicrobial agent.
 - D) pathogenic enzyme activity.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.4

- 6) The component that enriches chocolate agar is
- A) chocolate.
 - B) methyl brown indicator dye.
 - C) heat-lysed blood cells.
 - D) None of the answers are correct.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.3

- 7) Modified Thayer-Martin (MTM) agar
- A) incorporates antibiotics into the medium.
 - B) enhances the growth of normal flora.
 - C) prevents the growth of *Neisseria gonorrhoeae*, but promotes the growth of *Neisseria meningitidis*.
 - D) prevents the growth of *Neisseria meningitides*, but promotes the growth of *Neisseria gonorrhoeae*.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.3

- 8) The oxygen-free gas commonly employed during anoxic incubation of a bacterial culture is a mixture of
- A) nitrogen and carbon dioxide.
 - B) nitrogen and carbon monoxide.
 - C) nitrogen and argon.
 - D) nitrogen and hydrogen.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.4

9) Which type of medium allows a variety of microbes to grow but often changes color in the right conditions, such as when fermentation causes a pH change?

- A) selective but not differential
- B) differential but not selective
- C) both selective and differential
- D) neither selective nor differential

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.3

10) Some antibiotics inhibit protein synthesis by disruption of translation through interactions with the

- A) ribosome.
- B) protein.
- C) DNA.
- D) chromosome.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.10

11) The disk-diffusion test is used for

- A) assessing antimicrobial activity.
- B) the identification of *Escherichia coli*.
- C) the identification of viral plaques.
- D) the detection of antigen.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.4

12) Which of the following immunological tests would allow the specific identification of a microorganism directly in infected tissues?

- A) fluorescent antibody (FA)
- B) radioimmunoassay (RIA)
- C) enzyme-linked immunosorbent assay (EIA)
- D) agglutination

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.6

13) EMB agar preferentially selects for the growth of _____ bacteria.

- A) gram-positive
- B) gram-negative
- C) gram-variable
- D) both gram-positive and gram-negative

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.3

14) Latex bead agglutination tests are

A) rapid and reliable tests used for the detection of certain bacteria as well as certain serum antibodies.

B) inferior tests used in developing countries not having facilities for more reliable assays.

C) an integral part of the EIA test procedure.

D) too expensive for routine use.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.6

15) The first antibiotic to be characterized was a

A) cephalosporin.

B) quinolone.

C) β -lactam.

D) macrolide.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.10

16) One advantage of the fluorescent antibodies that permits disease diagnosis long before primary isolation techniques yield results is that

A) cross-reactions do not occur.

B) few controls are required.

C) nonspecific staining does not occur.

D) it may be applied directly to tissue.

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 28.6

17) Which of the following is a mechanism of aminoglycosides?

A) inhibition of cell wall biosynthesis

B) prevention of peptidoglycan crosslinking

C) inhibition of DNA replication

D) targeting the 30S ribosomal subunit

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.10

18) Which of the following would be useful for treating *Candida*?

A) an azole

B) a fusion inhibitor

C) a macrolide

D) a nucleoside reverse transcriptase inhibitor

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 28.11

19) The blotting of proteins from polyacrylamide gels to nitrocellulose paper, and the subsequent identification by specific antibodies, is called a(n) _____ blot.

- A) Western
- B) Eastern
- C) Southern
- D) Northern

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.7

20) Widespread antimicrobial drug resistance is usually passed by

- A) heterologous gene expression.
- B) reverse transcription.
- C) horizontal gene transfer.
- D) gene splicing.

Answer: C

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 28.12

21) Most drug-resistant bacteria isolated from patients contain a(n)

- A) R plasmid.
- B) T1 phage.
- C) reverse transcriptase.
- D) β -lactam.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.12

22) Antibody titer can be measured by

- A) negative microscopy.
- B) agglutination.
- C) plate counts done after isolating the pathogen.
- D) polymerase chain reaction.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.5

23) Compared with other precipitation tests, agglutination tests are

- A) more sensitive.
- B) less sensitive.
- C) about equally sensitive.
- D) highly variable in sensitivity, and thus the sensitivities cannot be compared.

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 28.6

24) Blood agar is an example of a medium that supports the growth of many microbes. Medium that supports a wide variety of growth is called _____ medium.

- A) selective
- B) general-purpose
- C) stable
- D) halophilic

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 27.3

25) Blood infections are most commonly caused by *Enterococcus* and _____, among many other genera.

- A) *Staphylococcus*
- B) Coryneform bacteria
- C) Propionibacteria
- D) *Escherichia coli*

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 28.3

26) Eukaryotic intestinal parasites, such as *Giardia*, can be identified by observing stool samples for the presence of

- A) cysts.
- B) antibodies.
- C) LPS.
- D) blood.

Answer: A

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 27.3

27) EMB agar is an example of a _____ medium.

- A) selective
- B) differential
- C) selective and differential
- D) neither selective nor differential

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.4

28) A laboratory in which Ebola virus is studied would be classified as a BSL-_____ laboratory.

- A) 1
- B) 2
- C) 3
- D) 4

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.1

29) The tuberculin skin test looks to identify _____ specific to *Mycobacterium tuberculosis*.

- A) antibodies
- B) antigens
- C) delayed hypersensitive (DTH) TH1 cells
- D) delayed hypersensitive (DTH) TC cells

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.5

30) Which of the following is a cause of drug-specific resistance in disease-causing organisms?

- A) inappropriate hygiene
- B) nosocomial infection
- C) poor immunity
- D) indiscriminate nonmedical use of antimicrobials

Answer: D

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 28.12

31) Which of the following is a novel antimicrobial target?

- A) disruption of lipid biosynthesis
- B) inhibition of protein synthesis
- C) inhibition of cell wall biosynthesis
- D) inhibition of peptidoglycan crosslinking

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.12

32) The basis for blood typing is referred to as

- A) neutralization.
- B) hemagglutination.
- C) immunodiffusion.
- D) precipitation.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.6

33) Which bacterium shows a characteristic metallic green sheen when grown on eosin-methylene blue (EMB) agar?

- A) *Escherichia coli*
- B) *Salmonella enterica*
- C) *Staphylococcus aureus*
- D) *Streptococcus pyogenes*

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.3

34) Protein A, commonly used in immunoblots, is derived from

- A) *Staphylococcus*.
- B) *Escherichia*.
- C) *Bacillus*.
- D) *Clostridium*.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.3

35) Two common or important bacterial pathogens found in the blood are

- A) *Pseudomonas aeruginosa* and *Escherichia coli*.
- B) *Enterobacter* and *Proteus*.
- C) *Shigella* and *Salmonella*.
- D) HIV and *Escherichia coli*.

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 28.3

36) A gram-negative, oxidase-positive diplococcus growing on chocolate agar likely belongs to the genus

- A) *Neisseria*.
- B) *Escherichia*.
- C) *Mycobacterium*.
- D) *Pseudomonas*.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.3

37) _____, periodic reports on the susceptibility of clinically isolated organisms to the antibiotics in current local use, are particularly valuable for tracking the emergence of new antibiotic-resistant strains of pathogens.

- A) Antibiotic susceptibility plates
- B) Antibiograms
- C) Direct agglutination reports
- D) Histograms

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.4

38) _____ defines the lowest quantity of an antigen that can be detected in a serological test.

- A) Specificity
- B) Antigenic capacity
- C) Minimal inhibitory concentration
- D) Sensitivity

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.5

39) Which of the following prevents supercoiling of bacterial DNA?

- A) macrolides
- B) quinolones
- C) penicillins
- D) cephalosporins

Answer: B

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 28.10

40) In the United States, every clinical and research institution requires that workers be protected from _____, a pathogen that is contracted almost exclusively from handling patient specimens (rather than from known exposure or accidents).

- A) anthrax
- B) hepatitis B
- C) drug-resistant mycobacteria
- D) influenza

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.1

41) The use of special culture media and incubation conditions to isolate microorganisms from samples is known as a(n)

- A) differential culture.
- B) selective culture.
- C) enrichment culture.
- D) sensitive culture.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.3

42) How are infections caused by the drug-resistant *Neisseria gonorrhoeae* treated?

- A) with β -lactamase-resistant antibiotics
- B) with tetracycline
- C) with synthetic analogs of tetracycline
- D) with vancomycin

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.10

43) An important intestinal pathogen generally acquired from contaminated food or water is

- A) *Clostridium perfringens*.
- B) *Mycobacterium tuberculosis*.
- C) Ebola virus.
- D) *Escherichia coli* O157:H7.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.3

44) Obligately anaerobic normal flora are found in _____, which is/are an anoxic habitat(s) of the body.

- A) the kidneys
- B) the skin
- C) portions of the oral cavity
- D) the lungs

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.3

45) A positive antibody titer indicates

- A) previous exposure to an antigen.
- B) active infection.
- C) latent disease.
- D) a healthy individual.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.5

46) Methylene blue in EMB agar slightly inhibits

- A) gram-positives.
- B) gram-negatives.
- C) viral plaques.
- D) both gram-positives and gram-negatives.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.3

47) The study of antigen-antibody reactions is known as

- A) serology.
- B) immunology.
- C) toxicology.
- D) epidemiology.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.5

48) Indirect EIA is used to detect

- A) antigens.
- B) antibodies.
- C) viruses.
- D) serum.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.7

49) Protein A has a strong affinity for

- A) antigen.
- B) blood.
- C) protein.
- D) antibody.

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 28.3

50) _____ is used to monitor gene expression of a pathogen.

- A) Reverse transcription PCR
- B) Serology
- C) Antigen capture assay
- D) Qualitative PCR

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.8

51) Hybridization requires a nucleic acid probe that binds to DNA or RNA of interest, which must be in the _____ form for hybridization to occur.

- A) single-stranded
- B) double-stranded
- C) oligomerized
- D) polymerized

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 28.8

52) Which of the following antibiotics inhibits RNA synthesis?

- A) penicillin
- B) erythromycin
- C) rifampin
- D) azithromycin

Answer: C

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 28.12

53) A mechanism for penicillin resistance in bacteria is

- A) acetylation of the antibiotic.
- B) phosphorylation of the antibiotic.
- C) confirmation change of the antibiotic.
- D) splitting the β -lactam ring of the antibiotic.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.12

54) SYBR Green, commonly used in qPCR, binds nonspecifically to _____, indicating the presence of the PCR product.

- A) ssDNA
- B) dsDNA
- C) ssRNA
- D) dsRNA

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.9

55) An antigen must be at least _____ amino acids long to be effective in a vaccine.

- A) 5
- B) 10
- C) 20
- D) 100

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.9

56) New vaccine strategies resulting from progress in biochemistry and molecular biology include the production of

- A) DNA vaccines.
- B) recombinant vector vaccines.
- C) synthetic peptide vaccines.
- D) DNA, recombinant vector, and synthetic peptide vaccines.

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 28.9

57) Which of the following immunogens is most effective as a vaccine?

- A) chemically inactivated bacteria
- B) dead virus
- C) live virus
- D) toxoid

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.9

58) An attenuated strain of a pathogen

- A) has lost its virulence, but may be used to create a vaccine.
- B) has changed its antigens to avoid an adaptive immune response.
- C) is more virulent than the regular strains and may cause septic shock.
- D) is an extracellular pathogen that will be cleared by innate immunity.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.9

59) Which of the following can be used in creating a vaccine?

- A) attenuated bacteria
- B) inactivated viruses
- C) recombinant proteins
- D) attenuated bacteria, inactivated viruses, or recombinant proteins

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.9

60) Why are secondary or "booster" reimmunizations given?

- A) Secondary immunizations are necessary for an innate immune response.
- B) Secondary reimmunizations produce a secondary immune response and boost antibody titers.
- C) Secondary reimmunizations produce a phagocytic immune response that is longer lasting than the primary immune response.
- D) Frequent secondary reimmunizations are needed because the immune system doesn't remember antigens for more than a few years.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.9

61) What is the drawback of live attenuated vaccines?

- A) They require more secondary reimmunizations than other vaccines.
- B) They are not effective in adults.
- C) They do NOT provide a long-lasting secondary immune response.
- D) They can cause disease in some immunocompromised individuals.

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 28.9

28.2 True/False Questions

1) Vaccines can be developed from synthetic peptides.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.9

2) A recombinant-vector virus would still be effective if the vaccinia virus is unable to express the cloned gene as an antigenic protein.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.9

3) DNA vaccines are bacterial plasmids that contain cloned DNA with an antigen of interest are unlike attenuated vaccines because there is no chance of causing disease.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.9

4) The causal agents of urinary tract diseases are often members of the normal flora.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.3

5) In cases of suspected foodborne or waterborne infections, fecal samples should be inoculated into a variety of selective media.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.3

6) Hydrogen sulfide production is assayed by growth in a medium containing ferric iron (Fe^{3+}).

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.3

7) The EIA test for HIV infection is a direct test for HIV envelope proteins.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.7

8) In the EIA test for HIV infection, the microtiter plate is coated with disrupted HIV particles.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.7

9) Many pathogens can be readily grown in laboratory culture.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.3

10) All positive HIV-EIA tests must be confirmed by another independent test, usually the HIV Western blot.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.7

11) *Salmonella* is commonly associated with wound infections.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.3

12) Gram stain smears of urethral exudates for the detection of *Neisseria gonorrhoeae* are highly specific.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.3

13) Nucleic acid probes are used to detect the presence of particular sequences of DNA.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.8

14) One problem with the EIA test is the disposal of radioactive wastes.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.7

15) When using the EIA test for HIV screening, BOTH false negative and false positive results are possible.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.7

16) Chitin is found ONLY in fungi and insects.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.11

17) At this time, EIA tests are too difficult for routine diagnoses.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.7

18) Nucleic acid probes and immunological reagents are NOT commonly used in the clinical laboratory for the diagnosis of viral disease, because the cultivation of the virus is often difficult or even impossible.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.8

19) Some infectious agents, such as *Neisseria gonorrhoeae*, do NOT elicit a systemic immune response.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.5

20) Fluorescent antibodies can be used to separate mixtures of cells into relatively pure populations with a fluorescence cytometer.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.6

21) To detect antibodies in human serum, the direct EIA is most commonly employed.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.7

22) Literally hundreds of clinically useful EIAs have been developed.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.7

23) Interferons promote viral replication.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.11

24) A toxoid is an attenuated form of a toxin that retains both its antigenicity and toxicity.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.9

28.3 Essay Questions

1) Use specific examples to describe the difference between a differential and a selective medium.

Answer: A selective medium restricts the growth to a particular phenotype of interest. For example, a penicillin-containing medium could be used to select for microorganisms resistant to penicillin. Halophilic microbes could also be selected for by including a high concentration of salt to the medium, which would ensure non-halophiles could not grow on the selective medium. A differential medium does not restrict growth of particular organisms but instead permits the identification of a particular phenotype. For example, acid production used as a proxy for sugar fermentation can be observed through the addition of a pH-indicating dye into the medium.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.3

2) Describe the use of combination drugs and why they are effective. Provide examples.

Answer: The efficacy of some antibiotics can be retained if they are given with compounds that inhibit antibiotic resistance. Several β -lactam antibiotics can be combined with β -lactamase resistant microorganisms. For example, ampicillin can be mixed with a β -lactamase inhibitor that binds irreversibly, preventing degradation of the ampicillin. Also, the use of sulfamethoxazole-trimethoprim prevents the loss of efficacy through mutation and selection of resistance.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 28.12

3) Describe the three main components to PCR-based testing.

Answer: The first step requires that nucleic acids be extracted from the sample, for example through bead-beating cell lysis. Amplification of the DNA or RNA is then performed using primers specifically targeting the nucleic acid sequence(s) of interest. Finally, observation of the PCR product can be done either with or without enhanced visualization systems, which generally first required the PCR amplicons to be electrophoresed.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.8

4) Propose a series of procedures to determine whether a patient has a urinary tract infection.

Answer: Direct microscopic examination of urine may indicate the presence of abnormal numbers of bacteria in the urine. Also, a Gram stain may be done on urine samples to identify the morphology of potential urinary tract pathogens. Blood agar can be useful for initial isolation. Selective and differential enteric media such as eosin-methylene blue agar permit the initial differentiation of lactose fermenters from non-lactose fermenters while inhibiting the growth of gram-positives. Urine cultures can be done quantitatively by counting colonies on either blood agar or a selective and differential growth medium.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 28.3

5) Describe mechanisms of transfer of healthcare-associated infections (HAIs).

Answer: Some HAIs are acquired from patients with communicable diseases, but others are selected and maintained within the hospital environment. HAIs are often present as normal flora in either patients or healthcare staff. HAIs are acquired because patients, such as newborns and the elderly, are already ill or immunocompromised. Furthermore, healthcare personnel can transfer pathogens between and among patients. Surgery, catheterization, and intravenous lines compromise the skin and mucus membranes (potentially introducing pathogens into the body). Anti-inflammatory drugs and antibiotic treatment may select for resistant and opportunistic pathogens as well.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 28.2

6) Discuss how recent advances in the molecular biology of nucleic acids have revolutionized methods used for the identification of infectious disease agents and how they are favored over traditional biochemical tests.

Answer: Molecular biology techniques that generally look to identify gene sequences or mRNA transcripts are especially powerful in clinical diagnostics, because they do not usually require propagation of cells and culturing techniques that can take days. Biochemical tests can reveal only the presence/absence, whereas molecular techniques such as real-time PCR can provide information about the quantity of a particular taxon.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 28.8

7) Differentiate between the various biosafety levels and when they are most appropriately used.

Answer: Biosafety level 1 (BSL-1) is often used in teaching laboratories that do not work with known pathogens. In this case, barrier protection (gloves, coat, safety glasses) is recommended for use. Biosafety level 2 (BSL-2) is appropriate for a laboratory that works with moderate-risk pathogens. Barrier protection must be used, and manipulations that may generate aerosols must be performed in a biological safety cabinet. Biosafety level 3 (BSL-3) is appropriate for a laboratory that works with emerging and high-risk pathogens; barrier protection must be used. The laboratory is also negatively pressurized and fitted with filters to prevent pathogen escape. A biological safety cabinet is also in place for pathogen manipulations. Biosafety level 4 (BSL-4) is appropriate for a laboratory that works with emerging and high-risk pathogens, especially those spread by aerosols, for which there is no treatment, cure, or vaccine; barrier protection must be worn. Manipulations in a biosafety level 4 laboratory must be performed in a sealed biological safety cabinet or by personnel wearing positive pressure suits fitted with an air supply.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 28.1

8) What is the basis for inferring infection of a pathogen when a serological EIA is performed? In this context, what does a positive result indicate?

Answer: An EIA can measure the titer of a specific antibody using a probe specific to the antibody of interest. A patient's serum should contain antibodies produced by the immune system if they were exposed to a particular pathogen or are currently infected by the pathogen. A positive result on this type of test suggests either situation is probable.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 28.7

9) Discuss antimicrobial susceptibility testing and the use of minimal inhibitory concentration (MIC) values.

Answer: Antimicrobial susceptibility is measured by determining the smallest amount of agent needed to completely inhibit the growth of the test organism *in vitro*. This value is called the minimum inhibitory concentration (MIC). The MIC is determined by preparing a series of culture tubes that are inoculated with the same number of microorganisms. Each tube contains liquid medium with increasing concentration of the antimicrobial agent. After incubation, the tubes are checked for visible growth. This is usually completed using an automated format. A miniaturized version of this test shows a standard microtiter method using two-fold dilutions of several antibiotics in medium inoculated with a standard amount of the test bacteria.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.4

10) Explain the principle difference between direct agglutination and passive agglutination.

Answer: Direct agglutination is clumping formed from soluble antibodies binding to soluble antigens that are either bound to the surface of a cell or bound to an insoluble particle. Passive agglutination again uses soluble antibodies, but the antigens of interest are also soluble, so they are first adsorbed to an inert antigen-carrier such as charcoal particles, latex beads, or red blood cells.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 28.5

11) When a fecal sample is collected from an ill patient, how are each of the following identified: bacterial cell, eukaryotic protist, and virus?

Answer: Bacterial cells can be cultured on selective media from a fecal sample, and biochemical tests can follow to identify which bacterial pathogen(s) are present. Eukaryotic protists and viruses are generally not propagated in a lab for clinical diagnostics. Pathogenic protists are often identified via microscopy. All three groups can be identified through molecular diagnostics as well, such as the use of Northern, Southern, and Western blots, real-time polymerase chain reaction (qPCR), and reverse transcriptase (RT)-PCR.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 28.11

12) Assume you are the procurement manager of the medical charity "Doctors Without Borders" and there was an imminent infectious disease epidemic threat in a remote village in Africa. The epidemic can be prevented by vaccinating the population of interest. However, due to bad roads and lack of fast transportation services to reach the site of interest, which type of vaccine would you purchase for this job considering the length of time it would take to reach the affected site and why? A live attenuated virus or killed virus vaccines?

Answer: Since attenuated viruses are difficult to select, standardize, and maintain and have a limited shelf life and require refrigeration, which may not be available due to the nature of the job at hand, it would be more advisable to go for a killed virus vaccine, which is relatively easy to store and maintains its potency for long periods of time.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 28.9

13) Why are rabies shots given to domestic animals yearly and not just once in the animal's lifetime?

Answer: After the initial rabies vaccine immunization, the antibody titer in an animal decreases over time, but subsequent yearly exposure to the same antigen causes another memory response. This annual shot is known as a "booster shot." Periodic reimmunization maintains high levels of memory B cells and circulating rabies antibodies, providing long term active protection against rabies diseases.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 28.9